Government Data APIs

Week 11 · APIs module · Textbook Chapter 11

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This week at a glance

Four public APIs, each showing one idea that holds for every API. This week, every class is live coding, from one lab notebook in three parts.

Monday | Lab, part 1: the Census

- Keys, and *what can I ask for?* — the variable and geography catalogs

Wednesday | Lab, part 2: FRED, then the EIA

- *Let the server compute*, then *let the server filter*

Friday | Lab, part 3: Congress.gov

- *Follow the links*; join two APIs; then textbook revisions

The lab

`labs/week-11/lab-11.ipynb` in the course repository. Each part is added before its day.

Four keys, as environment variables

`CENSUS_KEY` today; `FRED_KEY` and `EIA_KEY` Wednesday; `DATAGOV_KEY` Friday. Links are in the Canvas announcement and the lab's first cell.

Due Sunday, Nov 1

Chapter 11's **Recommended Exercises**, from its companion notebook.

1. Monday · Live-Coding Lab

This week, every class is live coding

Find it | `labs/` in the course repository

1. Open the **labs** folder of `cuinfoscience/INF04617-Fall2026` on GitHub
2. Open `week-11`, click `lab-11.ipynb`, then **Download raw file**
3. Open it in JupyterLab in the `webdata` environment, *after* setting your keys

Use it | one part a day

- I write the code on screen, and you write it in your copy
- Put **your** email in the User-Agent in Step 0
- Today: Part 1, Steps 0–6, about 50 minutes
- Wednesday and Friday: download the new copy and start at that part's setup step

Solutions

`lab-11-key.ipynb`, beside it: every cell filled in and run. Fall behind? Copy the cell from the key and keep going.

Nothing to submit

The lab is practice. After each step, a **Discuss** box holds questions we'll talk over before the next.

Error flavors

A missing key stops Step 0 *on purpose*: set it and run again. A cell that fails with your key set is *broken*: tell me, or open an issue.

No key yet?

Follow along: the key shows every output. A Census key works only after you click the link in its email.

Public data by mandate

Government APIs exist because of democratic commitments to **transparency** — not business strategy:

- The **Census Bureau** is constitutionally mandated to count the population.
- The **Federal Reserve** publishes the economic data it collects; FRED is the St. Louis Fed's service for it.
- The **Energy Information Administration** was created in 1977, after the energy crises, to publish independent energy statistics.
- The **Library of Congress** publishes Congress's legislative record at Congress.gov.

They embody the **openness** value from Ch. 3, *The Post-API Age*: public data, freely accessible by mandate rather than platform goodwill.

Stable, but less polished

Agencies don't pivot business models or get acquired — so these APIs are unusually **stable**. The tradeoff: dense docs, cryptic variable names, unhelpful errors. Navigating that *is* the skill.

Four APIs, four ideas

Week 10's Wikipedia taught **pagination**. Each API this week teaches one more idea you'll meet in every API:

Day	API	The idea
Monday	Census Bureau	**Discovery**: ask the API what you can ask for
Wednesday	FRED	**Let the server compute**: transforms in the request
Wednesday	EIA	**Let the server filter**: facets narrow the data before it's sent
Friday	Congress.gov	**Follow the links**: records point to related records

And one idea across all four: a **key** says who you are, and it stays out of your notebook.

Keys stay out of the notebook

A **key** identifies you, counts your requests, and enforces rate limits. A key typed into a notebook travels with it: into a commit, a submission, a classmate's laptop.

Set it in your shell *before* starting Jupyter, then read it:

```
# macOS/Linux: export CENSUS_KEY="your-key"
# Windows: setx CENSUS_KEY "your-key"
import os
census_key = os.environ["CENSUS_KEY"] # a missing key stops here
```

The User-Agent still says who sent each request: `WebDataScience/1.0 (your-email@colorado.edu)`.

The chapter's second way

A JSON file, `api_keys.json`, added to `.gitignore` *before your first commit*. A key in git history stays there even after you delete the file.

Names

This course uses `CENSUS_KEY`, `FRED_KEY`, `EIA_KEY`, and `DATAGOV_KEY`. Chapter 11 uses the same names.

Discovery: two catalogs before any data

What | the variable catalog

- Variables come in **groups**: `B25034` is "Year Structure Built"
- `B25034_002E`: group, line, and `E` for estimate (`M` is its margin of error)
- `.../acs5/groups/B25034.json` is itself an API

Where | the geography catalog

- `.../acs5/geography.json` lists every level the dataset answers for
- Each says what it **requires**: a `place` needs a `state`; a `tract` needs a `state` and a `county`
- Places are **FIPS codes**, as strings: Colorado `08`, Boulder `07850`

Codes change between survey years: learn to look them up, not to memorize them. Then **sanity-check** one value: Boulder's median household income is $87,493 (±$6,221), not 7 or 70 million.

Census — from the catalogs to a real place

Name the variables with `get`, the place with `for`, and what it requires with `in`:

```
census_base = "https://api.census.gov/data/2024/acs/acs5"
params = {"get": "NAME,B25034_001E,B25034_002E,B25034_003E",
          "for": "place:07850", # Boulder: a FIPS code, kept as a string
          "in": "state:08", # a place requires its state
          "key": census_key}
data = requests.get(census_base, params=params, headers=HEADERS).json()

df = pd.DataFrame(data[1:], columns=data[0]) # row 0 is the header
numeric_cols = [c for c in df.columns if c.startswith("B25034")]
df[numeric_cols] = df[numeric_cols].apply(pd.to_numeric) # values arrive as strings
```

Boulder has 47,465 housing units; 646 were built in 2020 or later, 3,836 in the 2010s.

When there's no estimate: `-666666666`

83 of Colorado's 482 places have no median-income estimate in 2020–2024: too few households answered. The API doesn't leave the value out — it sends a number that can't be an income.

Place	People	Median income
Boulder	106,433	87,493
Campo	58	53,125 (±40,478)
Bonanza	20	`-666666666`
Carbonate	0	`-666666666`

Average it as is

The mean of that column is −333,298,178. Mask the codes first: `towns.mask(towns < 0)`.

What the codes mean

The Census Bureau's notes on estimate and annotation values list each one; a margin of error of `-222222222` goes with it.

2. Wednesday · Live-Coding Lab

Today: Part 2, FRED and then the EIA

Both of today's APIs do work *before* they answer:

- **FRED** (Federal Reserve Economic Data, from the St. Louis Fed) **computes**: changes, percent changes, yearly averages
- The **EIA** (U.S. Energy Information Administration) **filters**: one state, one sector, one column, one span of years

Asking the server, instead of downloading everything and doing it in pandas, is today's idea.

Use it | Steps 7–14, about 50 minutes

- Download the new `lab-11.ipynb`; start at **Step 7**, which imports and reads the keys again
- Put **your** email in the User-Agent

Two keys today

`FRED_KEY` and `EIA_KEY`, set *before* you start Jupyter. Both APIs call the parameter `api_key`; Monday's Census called it `key`.

No key yet?

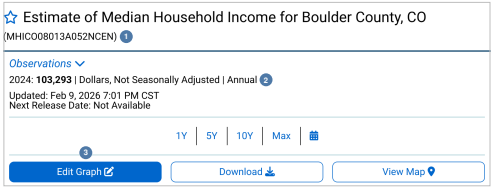
The EIA's arrives by email within minutes. Follow along from the key until it does.

FRED: find the series, then ask what it is

Every one of FRED's 800,000 series has an **ID** (1), on its page. The `series` endpoint describes it; `series/observations` sends its values:

```
fred_base = "https://api.stlouisfed.org/fred"
params = {"series_id": "MHICO08013A052NCEN",
          "api_key": fred_key, "file_type": "json",
          "observation_end": "2024-12-31"}
data = requests.get(f"{fred_base}/series/observations",
                    params=params, headers=HEADERS).json()
income = pd.DataFrame(data["observations"])[["date", "value"]]
income["value"] = pd.to_numeric(income["value"],
                                errors="coerce") # "." is missing
```

2024: $103,293 (2) — not the Census API's $103,994. Different programs: name the one you cite.



Boulder County's median household income on FRED, captured 2026-10-25 (chapter 11, figure 11.2).

Let the server compute: `units`

Step 9 gathers Step 8's lines into `fred_observations(series_id, **params)`. Then the *same series* twice more, one parameter changed:

```
change = fred_observations("MHICO08013A052NCEN",
           units="chg", observation_end="2024-12-31")
growth = fred_observations("MHICO08013A052NCEN",
           units="pch", observation_end="2024-12-31")
```

Year	Dollars	Change	Percent change
2022	96,584	6,257	6.9
2023	97,170	586	0.6
2024	103,293	6,123	6.3

What `units` can be

`lin`	as published
`chg`	change
`ch1`	change from a year before
`pch`	percent change
`pc1`	percent change from a year before
`log`	natural logarithm

Microsprint

In which years did the income *fall*? Six, from 2001 to 2020; the largest, $3,003 in 2009.

Let the server compute: `frequency`, then `units`

One series, `CPIAUCNS` (the national Consumer Price Index, monthly), three requests:

```
years = {"observation_start": "2015-01-01",
         "observation_end": "2025-12-31"}
monthly = fred_observations("CPIAUCNS", **years)
annual = fred_observations("CPIAUCNS", frequency="a",
           aggregation_method="avg", **years)
inflation = fred_observations("CPIAUCNS", frequency="a",
           aggregation_method="avg", units="pc1", **years)
```

FRED lowers the frequency first, then applies the units: each year's mean of twelve months, then its percent change from the year before.

Trust, but check once

Microsprint: the same numbers from `monthly`, with `groupby()` and `pct_change()`.

Year	FRED	pandas
2023	4.1	4.1
2024	2.9	2.9
2025	2.6	2.6

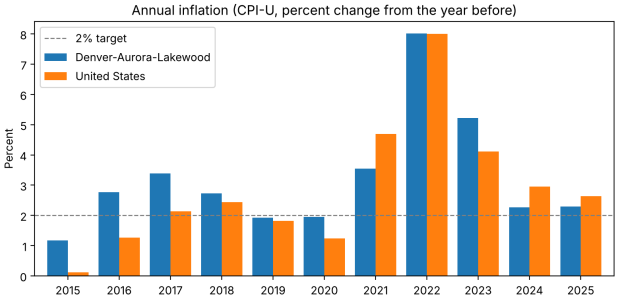
Why can FRED give 2015's inflation, and pandas can't?

Denver against the nation

The Bureau of Labor Statistics' CPI for the Denver area, `CUUSA433SA0`, is annual already: it needs only `units="pc1"`.

```
denver = fred_observations(
    "CUUSA433SA0", units="pc1", **years)
cpi = denver.merge(inflation, on="date",
    suffixes=("_denver", "_national"))
```

Denver ran ahead from 2015 to 2020, and behind since 2024. Housing is the largest part of the CPI: what would you ask FRED next?



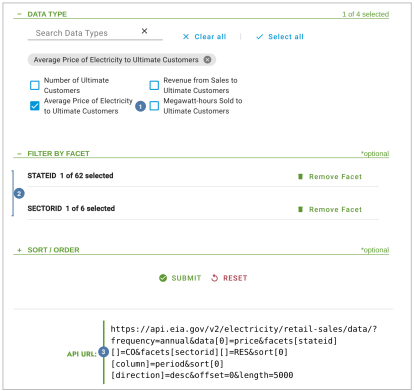
Drawn by the chapter's code from FRED, 2026-10-25 (chapter 11, figure 11.6). The dashed line is the Federal Reserve's 2% target.

EIA: routes that describe themselves

The EIA's API is a tree of **routes**. Ask for one without `data/`, and it says what it holds:

```
eia_base = "https://api.eia.gov/v2"
meta = requests.get(f"{eia_base}/electricity/retail-sales/",
                    params={"api_key": eia_key},
                    headers=HEADERS).json()["response"]
print([f["id"] for f in meta["facets"]])
# ['stateid', 'sectorid']
for column, about in meta["data"].items():
    print(column, "|", about["units"])
# price | cents per kilowatt-hour ...and 3 more
```

A **facet** filters the rows; a **data column** (1) is what each row holds. Monday's idea, *what can I ask for?*, from a different API.



The EIA's API Dashboard, captured 2026-10-25 (chapter 11, figure 11.3): its menus (2) are those routes, and it writes the request (3).

Let the server filter

Add `data/` to the route, and each parameter narrows the rows. The brackets are part of the names:

```
params = {"frequency": "annual", "data[]": "price",
          "facets[stateid][]": "CO",
          "facets[sectorid][]": "RES", # homes
          "start": "2001", "end": "2025",
          "sort[0][column]": "period",
          "sort[0][direction]": "asc",
          "api_key": eia_key}
prices = requests.get(f"{eia_base}/electricity/retail-sales/data/",
            params=params, headers=HEADERS).json()["response"]
print("Rows:", prices["total"]) # Rows: 25
```

25 rows, from a route that holds every state and sector by month, quarter, and year. Colorado's homes paid 15.69 cents a kilowatt-hour in 2025.

Microsprint

Change one facet: a sector (`COM`, `IND`) or a state (`UT`, `NM`). Colorado's businesses paid 12.37 cents in 2025.

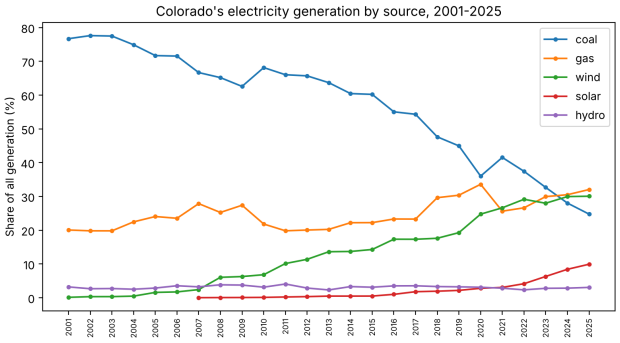
Still strings

Prices and `total` arrive as text, like the Census Bureau's numbers: `pd.to_numeric()` before any math.

What a filtered request can show

The chapter asks the same API for Colorado's annual generation from five sources, plus the total: one facet, six values, one request.

Coal made 76.7% of the state's electricity in 2001 and 24.7% in 2025. Wind grew from almost nothing to 30.0%; 2024 was the first year coal wasn't the largest source.



Drawn by the chapter's code from the EIA's API, 2026-10-25 (chapter 11, figure 11.7).

Paging: `offset` and `length`, against Wikipedia's `continue`

EIA | you keep count

- At most 5,000 rows a request; `total` says how many match
- Colorado's monthly generation, 2001–2025: 9,289 rows
- Page two: `offset=5000`, the rows to skip
- Stop when you have `total` rows — the `"incomplete return"` warning comes with the last page too

Wikipedia | the server keeps count

- Week 10: each answer carried a `continue` token
- You sent it back without knowing what it meant
- Stop when there's no token

Sort when you page, so no row moves between pages. And **filter first**: a facet, a shorter span, or a lower frequency may fit the answer in one page.

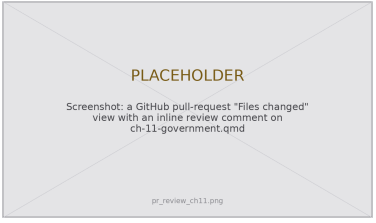
3. Friday · Live-Coding Lab

Improving Chapter 11 together

Today we make the book better. Each of you opens a **pull request** against `ch-11-government.qmd` and reviews a classmate's.

The workflow (from Week 1):

1. Branch / fork the book repo
2. Edit the chapter; commit with a clear message
3. Open a PR describing *what* and *why*
4. Review two peers' PRs: kind, specific, actionable



A revision menu for Chapter 11

High-value targets — pick one and scope it to a single PR:

- **Test the code against live APIs.** Series IDs, endpoints, and election years drift; keys expire. Run the notebook, report what broke, and fix it.
- **Close a gap in "Common Issues to Debug."** Add a worked `-666666666` sentinel-value example, or a rate-limit / retry snippet (FRED ≈120 req/min; api.data.gov 1,000 req/hr).
- **Add a figure.** An annotated variable-code diagram, or the key → request → JSON → DataFrame pipeline.
- **Strengthen a thin exercise.** The six exercises are open-ended — add expected output, a rubric, or a starter cell so they self-check.
- **Extend "Further Reading."** A data-journalism example tied to the **Public Interest** callout — a documented withdrawn dataset or deprecated API.

A good revision, and a good review

A strong PR

- One focused change, clearly titled
- Explains the problem it fixes
- Builds (`quarto render`) without errors
- Matches the book's voice and notation
- Discloses any AI assistance

A strong review

- Runs / reads the change, not just skims
- Names specifics (line, claim, code)
- Separates "must fix" from "nice to have"
- Is generous — assume good faith
- Ends with a clear verdict

Next week: **Social & media platform APIs —** OAuth, Mastodon, and Bluesky in the post-API age.

Key takeaways

1. Government APIs serve **transparency mandates** — valuable, reliable, and unusually stable data.
2. The core skill is **navigating catalogs**, not memorizing codes that change every survey year.
3. **Guard your keys**: environment variables or a git-ignored config — never hardcoded.
4. **Combine sources** for stories no single dataset tells: Census says *what*, FRED says *why*.
5. Mind the **institutional context** — using open data well is a form of civic engagement.